7th Grade Mathematics Exam

Name: ______________________________ Date: _______________

Time: 60 minutes | Total: 60 points

Show all work. Calculators are not permitted.

Part A — Proportionality & Proportion Coefficient (20 pts)

Q1. (4 pts) The table below shows the distance traveled by a car at constant speed.

Time (h)	2	3	5	7
Distance (km)	70	105	175	245

a) Is this a proportional relationship? Justify.

b) Determine the proportion coefficient k.

c) Write a formula relating distance d to time t.

Q2. (4 pts) The table below is proportional with coefficient $k = 3$. Complete the missing values:

x	2	4	?	10
y	6	?	21	?

Q3. (4 pts) Solve for x using cross-multiplication: $4 / 10 = x / 25$.

Q4. (4 pts) A recipe uses **150 g of flour** for **4 cookies**. How many grams of flour are needed for **18 cookies**?

Q5. (4 pts) For each table, say whether the relationship is proportional. Justify.

Table a)

x	1	2	3	4
y	3	6	9	12

Table b)

x	1	2	3	4
y	3	4	5	6

Part B — Percentages (20 pts)

Q6. (3 pts) Calculate **25% of 80**.

Q7. (4 pts) In a class of 60 students, 15 wear glasses. What percentage of the class wears glasses?

Q8. (5 pts) A bicycle costs **200 €**. The price increases by **12%**. What is the new price?

Q9. (4 pts) A shirt originally priced **50 €** is discounted by **20%**. What is the final price?

Q10. (4 pts) Out of 250 students, 30% play a musical instrument. How many students play one?

Part C — Polynomials Basics (20 pts)

Let $P(x) = 3x^2 + 2x - 1$ and $Q(x) = x^2 - 5x + 4$.

Q11. (3 pts) Give the degree and the leading coefficient of $P(x)$.

Q12. (4 pts) Compute $P(x) + Q(x)$ and simplify.

Q13. (4 pts) Compute $P(x) - Q(x)$ and simplify.

Q14. (3 pts) Expand: $2x \cdot (x + 3)$.

Q15. (3 pts) Expand: $(x + 4)(x - 2)$.

Q16. (3 pts) Evaluate $P(2)$ where $P(x) = x^2 - 3x + 5$.

Bonus (+3 pts)

A jacket costs **80 €**. It is first reduced by **25%**, then the reduced price is increased by **25%**. Is the final price equal to 80 €? Justify with a calculation.